In [1]:
```python
import pandas as pd
```

In [11]:
```python
df = pd.read_excel(r"C:\Users\DELL\OneDrive\Desktop\jupyter notebook\customer-Purch
print(df.head())
```

```
  CustomerID  Product PurchaseDate  Quantity  UnitPrice    CustomerName  \
0      C5361    Phone   2024-03-05         8     618.83  Customer C5361
1      C6231   Laptop   2025-06-21         7     366.22  Customer C6231
2      C7704    Chair   2023-06-25         5     634.51  Customer C7704
3      C2923  Printer   2023-09-30         3     508.63  Customer C2923
4      C4847  Monitor   2023-04-03         4     452.06  Customer C4847

   ProductCategory PaymentMethod  ReviewRating  TotalPrice
0  Office Supplies          Cash             1     4950.64
1      Electronics    Debit Card             3     2563.54
2  Office Supplies   Credit Card             4     3172.55
3  Office Supplies     Gift Card             1     1525.89
4      Electronics   Credit Card             2     1808.24
```

In [12]:
```python
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1800 entries, 0 to 1799
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       1800 non-null   object
 1   Product          1800 non-null   object
 2   PurchaseDate     1800 non-null   datetime64[ns]
 3   Quantity         1800 non-null   int64
 4   UnitPrice        1800 non-null   float64
 5   CustomerName     1800 non-null   object
 6   ProductCategory  1800 non-null   object
 7   PaymentMethod    1800 non-null   object
 8   ReviewRating     1800 non-null   int64
 9   TotalPrice       1800 non-null   float64
dtypes: datetime64[ns](1), float64(2), int64(2), object(5)
memory usage: 140.8+ KB
None
```

In [13]:
```python
print(df.describe())
```

```
                PurchaseDate     Quantity    UnitPrice  ReviewRating  \
count                   1800  1800.000000  1800.000000   1800.000000
mean    2024-04-09 12:35:12     4.588889   395.729417      2.995000
min     2023-01-01 00:00:00     1.000000     5.180000      1.000000
25%     2023-08-22 00:00:00     3.000000   199.512500      2.000000
50%     2024-04-19 00:00:00     5.000000   391.970000      3.000000
75%     2024-11-25 06:00:00     7.000000   587.707500      4.000000
max     2025-06-30 00:00:00     8.000000   799.910000      5.000000
std                     NaN     2.313383   229.355902      1.425753

         TotalPrice
count  1800.000000
mean   1814.809333
min       6.060000
25%     585.320000
50%    1440.740000
75%    2795.300000
max    6389.120000
std    1473.605942
```

In [16]:
```
df = df.drop_duplicates()
```

In [17]:
```
df = df.dropna()
```

In [20]:
```
total_revenue = df["TotalPrice"].sum()
print("Total Revenue:",total_revenue)
```

```
Total Revenue: 3266656.8
```

In [31]:
```
category_sales = df.groupby("ProductCategory")["TotalPrice"].sum()
print(category_sales)
```

```
ProductCategory
Electronics        1049911.98
Furniture          1083282.25
Office Supplies    1133462.57
Name: TotalPrice, dtype: float64
```

In [33]:
```
top_customers = df.groupby("CustomerID")["TotalPrice"].sum().sort_values(ascending=
print(top_customers)
```

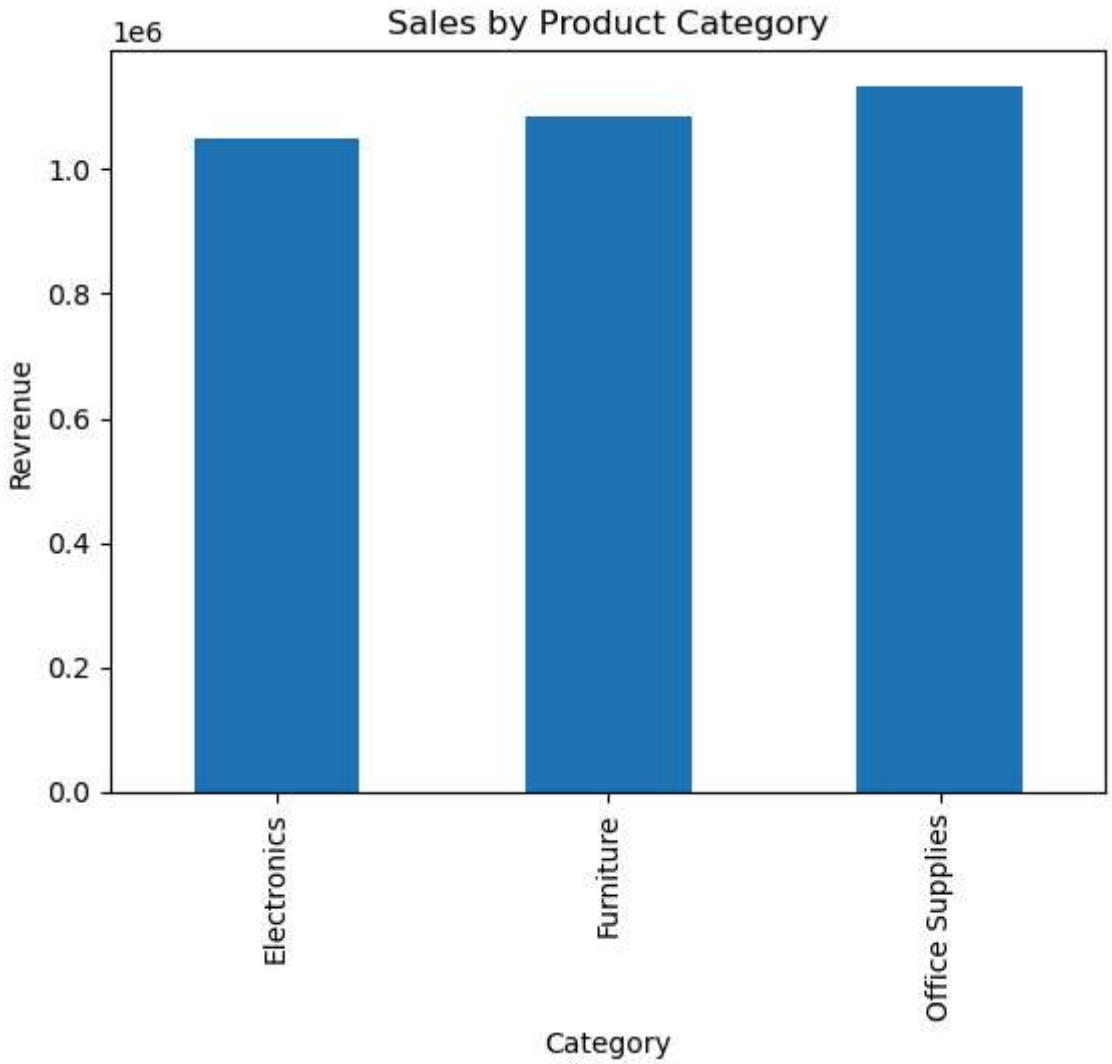
```
CustomerID
C3269    12172.80
C3460     9060.66
C3979     8386.73
C1528     7815.36
C9618     7754.45
Name: TotalPrice, dtype: float64
```

In [34]:
```
payment_sales =df.groupby("PaymentMethod")["TotalPrice"].sum()
print(payment_sales)
```

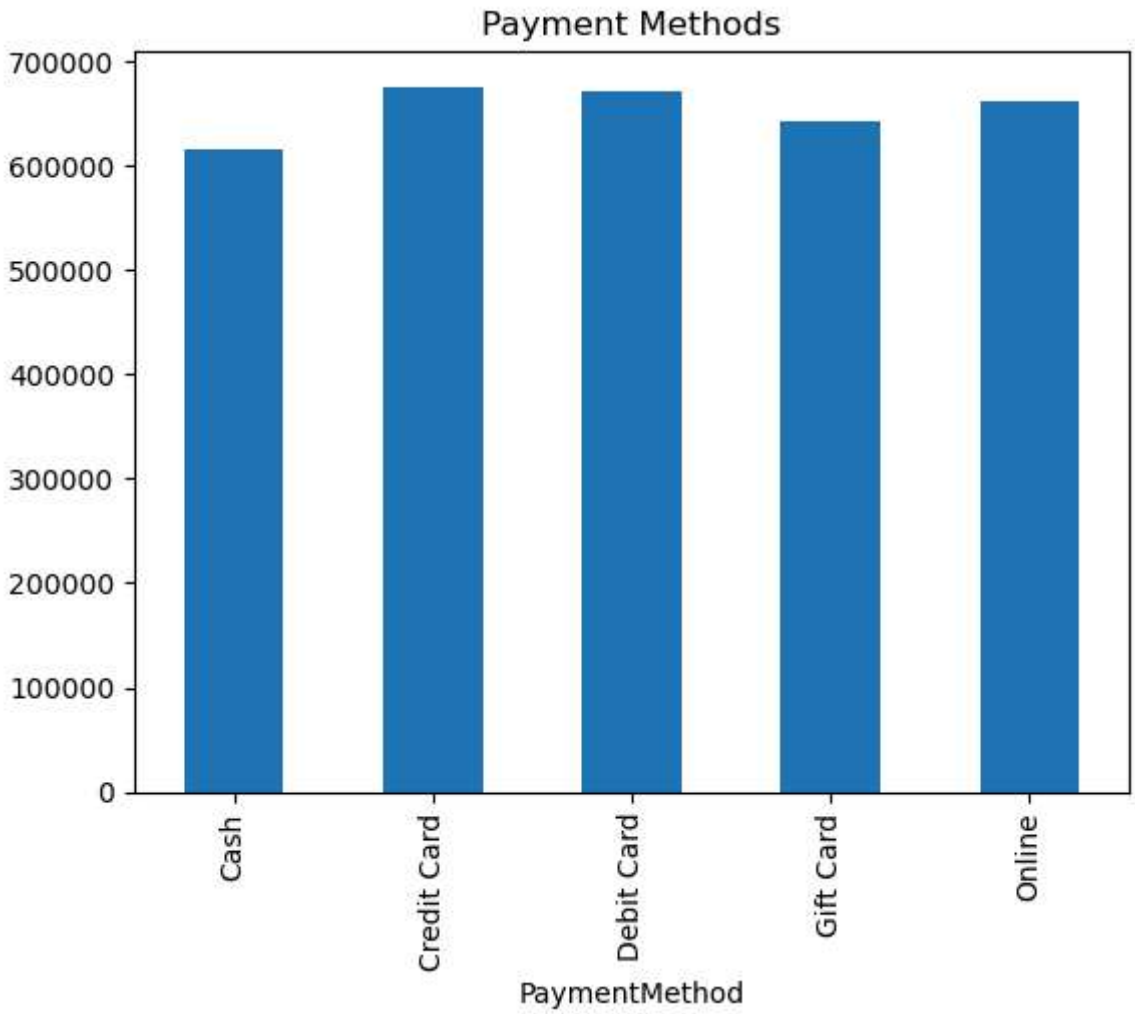
```
PaymentMethod
Cash           615791.82
Credit Card    675771.51
Debit Card     671304.23
Gift Card      642235.87
Online         661553.37
Name: TotalPrice, dtype: float64
```

In [35]:
```
import matplotlib.pyplot as plt
```

In [40]:
```
category_sales.plot(kind="bar")
plt.title("Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Revrenue")
plt.show()
```



In [42]:
```
payment_sales.plot(kind="bar")
plt.title("Payment Methods")
plt.show()
```


Payment Methods
700000
600000
500000
400000
300000
200000
100000
0
Cash
Credit Card
Debit Card
Gift Card
Online
PaymentMethod

In [ ]: